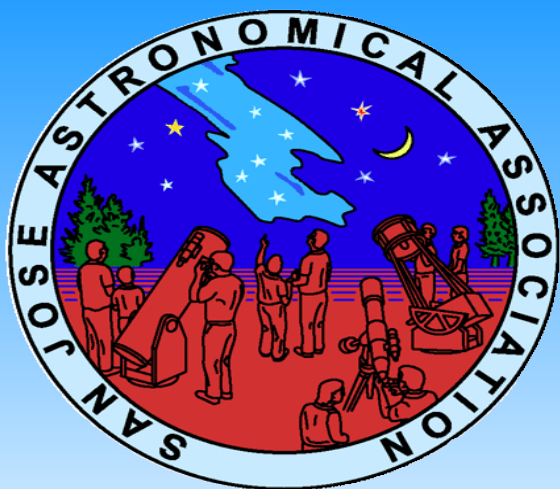




EPHEMERIS

September 2012

SJAA Activities Calendar

Jim Van Nuland

August (late)

- 24 Houge Park star party. Sunset 7:47 p.m. Star party hours: 8:45 until 11:45 p.m.
- 24 Imaging SIG meeting: 7:30 - 9:00 p.m. Houge Park. The topic is "DSLR or CCD"

p.m., 43% moon sets 11:20 p.m. Star party hours: 8:00 until 11:00 p.m.

- 29 **General Meeting.** Board meeting at 6:00; Social Time at 7:30; General Meeting at 8:00. Our speaker is Steve Gottlieb, speaking on Lord Rosse's Discovery of Spiral Nebulae.

September

- 1 **General Meeting.** Board meeting at 6:00; Social Time at 7:30; General Meeting at 8:00. Slide and Equipment Night.
- 2 Observing H-alpha flares and sunspots at Houge Park. Also our Telescope Tune-up Time. Sun party and tune-up hours: 2:00 - 4:00 p.m.
- 7 Astronomy Class at Houge Park. 7:30 p.m. The topic: Deep sky observing - galaxies, nebulae, clusters, etc.
- 7 Houge Park star party. Sunset 7:27 p.m., 53% moon rises 11:36 p.m. Star party hours: 8:30 - 11:30 p.m.
- 8 Dark-Sky weekend. Sunset 7:25 p.m., 44% moon rises 12:25 a.m.
- 15 Dark-Sky weekend. Sunset 7:14 p.m., no moon. Henry Coe Park's "Astronomy" lot has been reserved.
- 21 Houge Park star party. Sunset 7:05

October

- 5 Astronomy Class at Houge Park. 7:30 p.m. The topic: Fall Constellations / Highlight Objects. (outdoors)
- 5 Houge Park star party. Sunset 6:44 p.m., 70% moon rises 10:18 p.m. Star party hours: 7:45 until 10:45 p.m.
- 7 Observing H-alpha flares and sunspots at Houge Park. Also our Telescope Tune-up Time. Sun party and tune-up hours: 2:00 until 4:00 p.m.
- 13 Dark-Sky weekend. Sunset 6:33 p.m., no moon. Henry Coe Park's "Astronomy" lot has been reserved.
- 19 Houge Park star party. Sunset 6:25 p.m., 28% moon sets 10:14 p.m. Star party hours: 7:15 until 10:15 p.m.
- 27 **General Meeting.** Board meeting at 6:00; Social Time at 7:30; General Meeting at 8:00. Our speaker is Dr. Dana Backman, NASA, SOFIA. Specific topic to be announced.

A Deeper Look at Sunspots

Michael Packer

The year was 1769 in the highland country of Scotland that an astronomer first noticed a lowland feature on the solar disk. A cavity or hole in the Sun! What was the depression? It was the black core or umbra of your typical sunspot. Sunspots are in fact optical sinkholes where we see down about a thousand kilometers lower than the surrounding photosphere. The density of matter (H^+ ions) within a sunspot is also lower giving credence that it is more than just an optical hole. Why sunspots contain less material and are dark will be explained later but this hollowed aspect of sunspots can indeed be seen using a standard solar filter. The sunspot should have a decent sized umbra and penumbra and be observed as it approaches the edge of the solar disk. The depression is most noticeable at the solar limb where, instead of the umbra peaking above the surrounding penumbra, it's gone or mitigated. You're seeing right through it. The disappearance of the umbra near the limb is known as the Wilson Effect named after the Scottish astronomer Alexander Wilson who noted it.

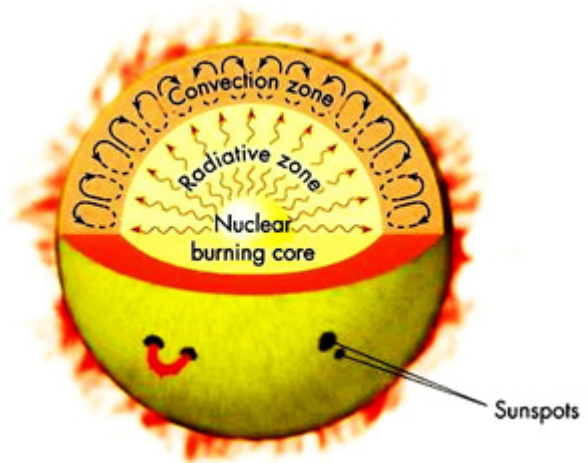
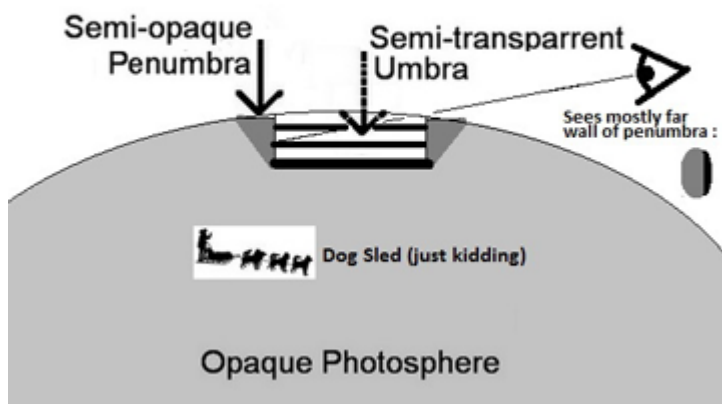
But can you look down the transparent hole? Can you get a better sense there is a depression? If your local astronomy club has access to an H-Alpha scope

24 hour news and information hotline:
(408) 559-1221
<http://www.sjaa.net>

Continued on page 2

<smile> the answer tilts to possible. In H-Alpha, the contrast and texture of the umbra, penumbra, granular photosphere and surrounding plage can be seen. Combined with oblique views of a good sized spot, this really makes 3D structure pop. It is in fact easy to see plage, bright arcs or “sinuous rivers” of white light, extend all the way down to “canyons” between granules of the photosphere. Sunspots themselves can look like a vortex of dark plasma precisely because they are a vortex of dark plasma. With binoviewers, a good sunspot group in H-Alpha can make you believe the Sun is going to eat itself inside out starting at the spot. (See video at <http://sanjoseastronomy.blogspot.com/2012/08/sun.html>) Just a few weeks ago I got a look of a really “sick” one. Do you recall 2010 Space Odyssey - the part where the obelisk begins to suck Jupiter’s outer cloud shell in? I suppose I could be exaggerating. But if you ask as Annabelle and Dale of Benicia who have never looked at the Sun, they easily called this dark abyss out for what it was. Without being told a thing they noted it and were repeating “whoa” like there was a team of dogsleds on the run inside the eyepiece... An H-Alpha Sunspot, it’s a magnetically cool phenomenon.

Sunspots Appear as a Depressions



Why Sunspots are dark: The photosphere is in some sense, the true surface of the Sun. Our eye sees it as a bright radiator of all visible light and it’s density is so high we cannot see through. It is opaque. Underneath lies the convection zone where energy (heat) is transported to the surface by the process of Hydrogen plasma (hydrogen atoms stripped of their electrons) churning (see cutout figure of sun). A churning gas of H ions is a positive current – just like electrons moving from one end of a wire to another is negative current. However a magnetic field can impede this current. Sunspots form when a section of this churning H ion gas is permeated by a magnetic field from the solar interior. The magnetic field strength can reach up to 4000 Gauss, 10,000 times the strength of Earth’s magnetic field. This magnetic force stops ion motion across it’s field lines and essentially quenches convection. Since convection is the main source of energy transport to the surface, less heat and radiant energy reaches the surface through the spot and the region looks significantly dimmer – dark. Sunspots are around 2000K cooler than the surrounding photosphere (5700K) because of the energy loss. And at this cooler temperature, recent studies have shown that H₂ molecules can form, align or couple with the magnetic field adding to the stability of the entire sunspot.

Why Sunspots are Semi-transparent: The opacity of the photosphere is due to H-ion density, which depends strongly on temperature. In sunspots since the temperature is lower the opacity is lower and we see deeper. In addition, the magnetic field force produces a pressure of its own inside sunspots. As the magnetic “B” field increases the pressure of the H ions inside the spot must fall to equalize with the surround photosphere (See equation). And if the pressure of the H ions falls, density is reduced again making the sunspot less opaque.

$$\frac{B_{\text{spot}}^2}{8\pi} + P_{\text{spot}} = P_{\text{surr}} \rightarrow P_{\text{spot}} \ll P_{\text{surr}} \rightarrow \rho_{\text{spot}} \ll \rho_{\text{surr}}$$

By similar use of the ideal gas law, the above equation also shows that if you increase the magnetic field strength within the convection zone, sunspots can grow in size/volume to keep the pressure equal to the surround. More on sunspot formation and viewing prominences in an upcoming issue.

Want to be instantly alerted when there is an active solar flare, or see a glossary of terms on the features of the sun? To learn more, check out some useful links here: <http://sanjoseastronomy.blogspot.com/2012/08/sun.html>.

Three Rings and an “Appulsar”

Akkana Peck

Last month was the annual NASA Lunar conference — a great chance to hobnob with lunar scientists and hear the latest on lunar research.

Orientele was big, with several talks on how its vast triple-ring structure formed. Orientele is my favorite lunar formation — a relatively young basin from a huge impact that raised three concentric rings of mountains. The outer ring, Montes Cordillera, is 930 km (580 miles) in diameter).

Orientele is also interesting to observers because it’s technically on the far side of the moon. We can only see it in profile, and only when the moon’s elliptical, inclined orbit gives it a tilt (called “libration”) that lets us see a little bit over the edge to the far side. (It’s probably best that it’s just over the edge. If we could see Orientele face on, like spacecraft do, it might look like a gigantic eye on the moon staring down at us, and human mythology and religion might have ended up quite different.)

It turns out there are several competing theories about exactly how the Orientele impact created its three rings, and about which mountain ring marked the edge of the original crater. Did material slumping back into a nested melt cavity cause fault blocks to rotate, creating mountains like a circular version of Nevada’s Basin and Range fault blocks? How big was the initial crater, and which mountain ranges were part of it? The GRAIL mission is measuring gravitational anomalies in the middle of the basin to try to pin down the details.

Happily for us, the structure is easily big enough to see with a small telescope. Since Orientele is on the far western edge of the moon, it’s best lit just before full moon and just before new moon. This month, the time around full moon isn’t a great time to look at Orientele —

it’s turned away from us then — but you can get a glimpse it in sunset light, just before the new moon, on the nights of Sep 13 and 14.

Interestingly, these multiple-ring basins we see on the moon (and even on Earth — like Upheaval Dome in Utah’s Canyonlands National Park) don’t exist on Mercury. There are basins with two rings, but any possible third rings are subtle and difficult to detect. Why? Nobody’s sure. Have they been covered by volcanic activity? Does the hotter rock make it react differently to impacts? Fascinating stuff.

Speaking of Mercury, you won’t see it this month — it’s too close to the sun. The other twilight planet, Venus, is in the morning sky. Meanwhile, evening twilight sports Saturn and Mars, both low and getting lower — Saturn will disappear in the glare before the month is out.

Jupiter rises around midnight and is a late night and early morning object, as are the outer planets. Pluto is well placed for viewing, and if you get frustrated looking for it you can always divert to M25 right next door, or nearby M18, M24, M22, M16 NGC6595 and a wealth of other deep sky objects. Neptune is just past opposition and should be easy to locate off the eastern tip of Capricornus (though it’s actually over the border in Aquarius).

Uranus reaches opposition right around midnight on the evening of Friday, Sep 28-29. But more interesting is what happens the previous Sunday, September 23rd, when Uranus passes within one arcminute of the star 44 Piscium.

Mark Gingrich, describing the upcoming event on the Shallow Sky list, coined the term “appulsar” for this fairly rare event. (An “appulse” is when two astronomical objects make a close approach.)

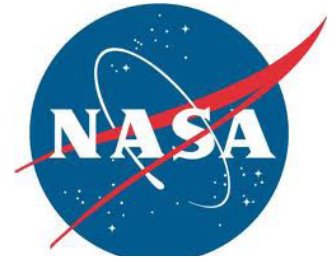
Both Uranus and the star are magnitude 5.75. Normally, each one is just barely visible with the naked eye from a fairly dark sky — a tough hunt for most of us from light polluted San Jose. With the two objects so close together, how much brighter will they look? Surprisingly, not that much brighter. Together, they’ll look like a single star around 5th magnitude. But that’s bright enough to see even in town — so it might be a novel chance to locate Uranus with the naked eye without leaving the Bay area.

Of course, it’s not just a naked eye event — it will be of interest to binocular and telescope users as well. One arc minute is a little bigger than Jupiter’s disk. So in binoculars, the pair should appear as a close double star — you may need to use a tripod or rest the binocular on a fence to separate the two. Can you tell, in binoculars, which one of the pair is the planet and which is the star? Uranus’s steady green glow, with no twinkling, should make it easy to tell which is which, at least if the air is at all steady. That’s my prediction, anyway. Let’s see if it turns out to be right.

Of course, in a telescope you’ll have no trouble telling the pair apart. But it should still be an interesting view. We know they’re almost exactly the same magnitude. So will a point source *look* the same brightness as a colored disk right next to it? Or will it look brighter?

Anyway, this should be a fun and unusual event, just a day after the Sep 22 autumnal equinox — a nice way to ring in the fall.





A Brand New Age: Queue Observing at Mt. Paranal

Dr. Marc J. Kuchner

First a caravan of white observatory cars arrives, winding up the narrow road to the 2600-m- (~8500-foot-) high summit. Then the shutters around the domes open, and rays from the setting sun alight on colossal mirrors and metal struts. It's the beginning of another busy night at Mt. Paranal, Chile, where I am learning about new, more efficient ways of managing a modern observatory.

I stepped into the observatory's control room to soak up some of the

new, unfamiliar culture. Here, under florescent lights and drop ceilings are banks of computer screens, one bank to control each of the four big telescopes on the mountaintop and a few others too. At each bank sits two people, a telescope operator and an astronomer.

The layout of this workspace was not unfamiliar to me. But the way these Mt. Paranal astronomers work certainly was. When I was cutting my teeth at Mt. Palomar observatory in California, I

would only go to the telescope to take my own data. In stark contrast, everyone observing at Mt. Paranal tonight is taking data for someone else.

The Mt. Paranal astronomers each spend 105 nights a year here on the mountain performing various duties, including taking data for other astronomers. The latter, they call "executing the queue." Headquarters in Germany decides what parts of the sky will have priority on any given night (the queue). Then



*European Southern Observatory
at Mt. Paranal, Chile.*

the Mt. Paranal astronomers march up the mountain and carry out this program, choosing calibrators, filling the log books, and adapting to changing conditions. They send the data back to headquarters, and from there it makes its way out to the wider astronomical community for study.

This new way of working allows the Mt. Paranal astronomers to specialize in just one or two telescope instruments each. Surely this plan is more efficient than the old-fashioned way, where each of us had to learn every instrument we used from scratch—sifting through manuals at 3:00 AM when the filter wheel got stuck or the cryogen ran out, watching precious observing time tick away. Here at Mt. Paranal, much of the work is done in a big room full of people, not off by yourself, reducing some dangers of the process. Also, queue observing cuts down on plane travel, an important step for cutting carbon emissions.

It's a brand new age, I thought as I watched the giant domes spin in the silent, cold Chilean night. And maybe with queue observing, some of the romance is gone. Still, my colleagues and I couldn't help saying as we stared out across the moonlit mountains: I can't believe how lucky we are to be here.

Dr. Marc J. Kuchner is an astrophysicist at the Exoplanets and Stellar Astrophysics Laboratory at NASA's Goddard Space Flight Center. NASA's Astrophysics Division works on big questions about the origin and evolution of the universe, galaxies, and planetary systems. Explore more at <http://www.science.nasa.gov/astrophysics/>. Kids can explore these topics at <http://spaceplace.nasa.gov/space>.

This article was provided by the Jet Propulsion Laboratory, California Institute of Technology, under a contract with the National Aeronautics and Space Administration.

Fixit Program

Ed Wong

The SJAA Fixit program provides service to club members and attendees by providing telescope repair, tune up and help in setting up telescope to use. We meet on the first Sunday 2pm-4pm of the month at the hall at Houge Park in conjunction with the SJAA Solar Observing program. This month we had a great turn out of folks who came to help and also those who brought scopes that need help. We worked on collimation of several scopes, provided instructions to several people who needed help understanding how to get the GOTO function of their scopes setup and working. We also helped some folks assemble a new scope they had just bought and also completed repairs to some of the scopes in the advanced scope loaner program. If you need help with some equipment, or you don't understand how to use it or it needs repair let us know and we will do our best to help you. Please let us know what you are bringing and we will try to get the right people set up to help you. There is a link from the main SJAA website under the "Fix it Day" section to assist you in doing this.



The SJAA Fixit program at work on August 5. Photo courtesy of Ed Wong.

ASTRONOMY magazine renewal time

Jim Van Nuland

It's time to renew our group subscription to Astronomy magazine. The rate for 2013 is still \$34, or \$60 for two years. Please send a check payable to me:

Jim Van Nuland,
3509 Calico Ave.,
San Jose CA 95124.

Subscribers: if I have your e-mail address, you should have gotten a note with particulars of your subscription. If not, write me, e-mail address below.

If you subscribe independently, and your subscription ends during 2012 or 13, you may convert to the group rate. Send a check and the renewal card or a mailing label to me, and you'll be added to the group for an additional 12/24 months.

If you do not subscribe and wish to do so, send the \$34/60 and your subscription will begin with the January 2012 issue.

I will hold your checks until late September, when the renewal package must be sent in. So don't worry that your check doesn't clear promptly.

Any questions? Call me at 408.371.1307, from 11 am to 11 pm, or e-mail to jvn@sjpc.org.

PLEASE NOTE: this applies to Astronomy magazine, not Sky & Telescope! The latter subscription is paid to the treasurer as part of your SJAA dues.

Good Reading!



The Last Month In Astronomy

AUG-05-2012 **Curiosity on Mars**

"We are back on Mars" is how JPL scientists described their feelings when the Mars Science Laboratory called "Curiosity" successfully landed. To do a number of new technologies had to work perfectly including the "sky crane" that lowered the rover to the surface. Among other things, there were 79 pyrotechnic charges that had to work. One "pfffft" and it was game over. <http://www.jpl.nasa.gov/msl/>

JUL-23-2012 **Sally Ride Dies**

America's First woman in apace, Sally K. Ride, has died from pancreatic cancer at age 61. She had been diagnosed at least 17 months earlier. Sally Ride rode on the Space Shuttle Challenger on June 18 1983. She also road on Challenger in 1984 and was scheduled to do so for a third time but the Challenger tragedy in January 1986 ended that possibility. She was an active member of both commisions following the shuttle disasters of 1986 and 2003. In 1987 Ride went to Stanford Univeristy at its Center for International Security and Arms Control. In 1989 she joined the faculty at U.C. San Diego. In 2001 she started the company Sally Ride Science creating science programs with an emphasis on getting the interest of young girls. <http://www.jpl.nasa.gov/news/news.cfm?release=2012-216>

JUL-18-2012 **Hot, Small and Close**

The Spitzer Infrared Space Observatory may have detected the smallest exoplanet discovered so far. The candidate planet, called UCF-1.01 is 33 light-years away and it appears to be two-thirds the size of Earth. The star that was being studied is GJ 436, already known to have a Neptunian sized planet. Spitzer, using a Kepler-style transit detection technique but in the infrared, detected unexpected dips. A look at archived data showed that the dips were periodic. The planet orbits its star once every 1.4 days and its temperature may be 1000 degrees Fahrenheit. <http://www.jpl.nasa.gov/news/news.cfm?release=2012-211>



It's a banner day at the SJAA. Photo courtesy of Ed Wong.

If You're Reading This - We Want You!

Mark Wagner

Don't worry, this is not military conscription or jury duty. Its just your humble astronomy club reaching out a friendly hand.

I'm Mark Wagner, President of SJAA through next February. I will be relieved, in more ways than one, to hand over the "reigns" of this starship to some capable and enthusiastic successor, a volunteer, who will help grow the club and find new ways to both offer new and exciting programs to our members and visitors, and to involve those with energy and ideas.

If you are reading this, that probably describes you!

Let me tell you about a few new programs, that are good examples of what may be done with the club. The foremost thing to realize is this is your club. I'll come back to that. So, what's new?

Solar Observing. We have a new solar telescope - a 100mm Lunt, and take it out for the public to look through from 2 to 4 PM every first Sunday of the month, at Houge Park. It is absolutely awesome. If you've never looked through a good solar scope, come out and see for yourself. This program is headed up by Michael Packer - a knowledgeable and enthusiastic solar observer (he owns an 80mm Lunt). We started this program just a few months back, and now club members are joining in bringing out their own solar setups for the public to enjoy. Sound fun? It is!

Fix-It Day. This is another new program, that is concurrent with Solar Sunday each month. From 2 to 4 PM we invite anyone with astro-gear they'd like looked at - to ascertain how to use it, or to repair it, to bring their goodies to Houge Park, where four volunteers (currently Ed Wong, Phil Chambers, Karl Ressinger and Dave Ittner) will provide man power and advice. So far we've had a good response, and the program is proving popular and effective.

Imaging Special Interest Group (SIG) - is headed up by Harsh Kaushikkar. It is still in its infancy, but the first few meetings have had a great turnout, and I'm certain with the popularity of imaging, that this will remain a vital program for the SJAA. You can learn more by contacting Harsh - he's also our Membership Chairman.

Quick STArT - headed up by new member Dave Ittner, with all sorts of creativity and enthusiasm - the club purchased several new Dobs and purged old gear from the loaner program, in order to offer beginner training on easy to use telescopes. Dave takes reservations by newbies, they get together as a group, and learn the basic ropes - then head out to a nearby darker site, where people get an eye full! New program, new ideas, new members, new fun for all! Kudos to Dave for coming up with this!

Social Time - - - huh? Social? Yes, we amateur astronomers actually can be sociable, despite our reputation for having a solitary hobby, we know getting together in groups is fun! So, SJAA has moved its board meeting back 1/2 an hour to allow some social time at 7:30 PM, before our General Meeting Speakers, so we can talk, have some refreshments, and get to know what everyone looks like (instead of just voices in the dark). It is fun!

The point is, this club is flexible. New things, new ideas, new blood, all are welcomed here. I could list other new endeavors as well, but the point would be the same - it is your club, and your ideas and enthusiasm that make things happen.

So, if you've read this - - - We Want You! Come to our Board Meetings - they are not "Bored Meetings" by any stretch. Your ideas, participation and enthusiasm are welcomed and you'll have an opportunity to have the club's resources behind you in your endeavor to share the hobby.

Here's an idea or two - we have a non-profit TV studio in San Jose, where we can produce our own programs. Anyone want to be the new Star Hustler? Or, we have people looking at land for an observatory, and others finding new local places to go observing. New ideas welcome here!

Check our schedule, and drop by the board meeting, or contact me. You won't be disappointed. Its a great group, and will welcome you.

Clear skies!

It Must Be Astronomical ...

Loaners

The telescope loaner program has been revamped. The program now includes QuickSTART, program geared to those new to astronomy. Please check it out at <http://www.sjaa.net/loaners.shtml>.

“The good thing about science is that it’s true whether or not you believe in it.”
- Neil deGrasse Tyson

School Star Parties

Completed Events					
	Total Sched.	Good Sky	Partial Success	Cloudy Fail	Cancel at noon
Jul	2	2			
Aug	-	-			
Sep	-	-			
Oct	-	-			
Nov	-	-			
Dec	-	-			
Total	2	2			
Scheduled - for the 2012/2013 school year					
	Total	Firm	Working		
Aug	1	1			
Sep					
Oct	3	2	1		
Nov	3	2	1		
Dec	1		1		
Total	8	5	3		

As of July 31, 2012

School Star Party Link

For information on school star parties including how to schedule one see <http://www.sjaa.net/school.shtml>.

Officers and Board of Directors

Pres Mark Wagner
VP Greg Claytor
Sec Rob Jaworski
Tres Robert Armstrong
Dir Lee Hoglan
Dir Rich Neuschaefer
Dir Rod Norden
Dir Kevin Roberts
Dir Michael Packer

Ephemeris Staff

Editors Paul & Mary Kohlmler
Circulation Mina Reyes-Wagner
Printing Accuprint (408) 287-7200

School Star Party Chairman

Jim Van Nuland (408) 371-1307

SJAA Email Addresses

Board of Directors sjaa-board@sjaa.net
 School Star Parties schools@sjaa.net
 Ephemeris ephemeris@sjaa.net

Other e-mail contacts are available at <http://www.sjaa.net/contacts.html>

Members Email Lists:
<http://www.sjaa.net/majordomo.html>

<http://sanjoseastronomy.blogspot.com/>
 twitter: [sj_astronomy](#)
<http://www.meetup.com/A-A-N-C/>

Publication Statement

SJAA Ephemeris, newsletter of the San Jose Astronomical Association, is published monthly.

San Jose Astronomical Association,
 P.O. Box 28243
 San Jose, CA 95159-8243

The SJAA Ephemeris is published in three formats: hardcopy, Adobe Acrobat PDF, and HTML. The PDF and HTML versions are found at <http://ephemeris.sjaa.net>.

Articles for publication should be submitted by the 10th of the previous month. The PDF version is generally available by the 24th of the previous month and the HTML version by the last day of the previous month.

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243

ADDRESS SERVICE REQUESTED

San Jose Astronomical Association Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

☐ **New** ☐ **Renewal** (Name only if no corrections)

Membership Type:

- ☐ Regular — \$20
☐ Regular with Sky & Telescope — \$53
☐ Junior (under 18) — \$10
☐ Junior with Sky & Telescope — \$43

Subscribing to Sky & Telescope magazine through the SJAA saves you \$5 off the regular rate. (S&T will not accept multi-year subscriptions through the club program. Allow 2 months lead time.)

☐ I prefer to get the Ephemeris newsletter in print form (Add \$10 to the dues listed on the left). The newsletter is always available online at <http://ephemeris.sjaa.net>
Questions?

Send e-mail to membership@sjaa.net

Bring this form to any SJAA Meeting or send to the club address (above). Please make checks payable to "SJAA".

You can join or renew online:
<http://www.sjaa.net/membership.shtml>

Name: _____

Address: _____

City/ST/Zip: _____

Phone: _____

E-mail address: _____